Price Transmission

from World Markets to Developing Countries' Domestic Markets

How global food price shocks reach farmers and consumers in the developing world

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Outline

The Presentation builds in three steps:

How does transmission work?

Why is it incomplete?

Who does it affect?

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2. The World Price & Spatial Arbitrage
3. Benchmark: the Law of One Price
4. Three Dimensions of Transmission
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Motivation: Why Does Price Transmission Matter?



Photo: BBC News Indonesia (May 2026)

When responding to Economists about the weakening of Indonesian currency rupiah towards American dollar.

"Villagers do not use dollars" – Prabowo Subianto, current President of Indonesia.

+83%

Global food price spike (2007–08)

~70%

Developing countries net food importers

40–75%

Household's budget goes to food (vs ~8% in the UK)

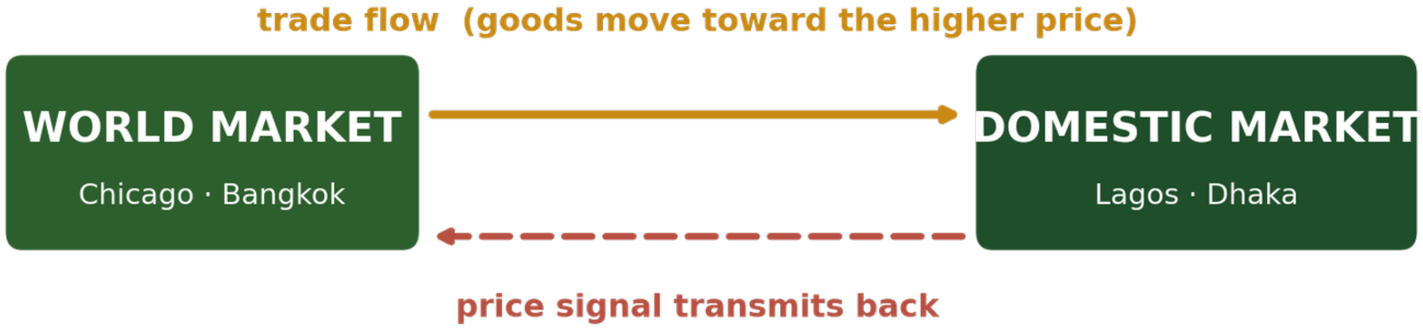
The Goal

How strongly, and how fast, do world commodity price changes pass through to domestic food prices in developing countries ?

Sources: FAO (2011); Headey & Fan (2008); Ivanic & Martin (2008)

The World Price & Spatial Arbitrage

A **world price** is a real benchmark price at a major hub — Chicago (wheat), Bangkok (rice), Rotterdam (oil).



If wheat is cheaper in Chicago than Lagos (after transport), traders buy low and sell high. This is **Arbitrage.**

Transmission asks: how much of a Chicago shock actually reaches Lagos?

> **Arbitrage is the mechanism behind all price transmission theory**

Sources: Fackler & Goodwin (2001); Rapsomanikis et al. (2006).

The Benchmark — Law of One Price (LOP)

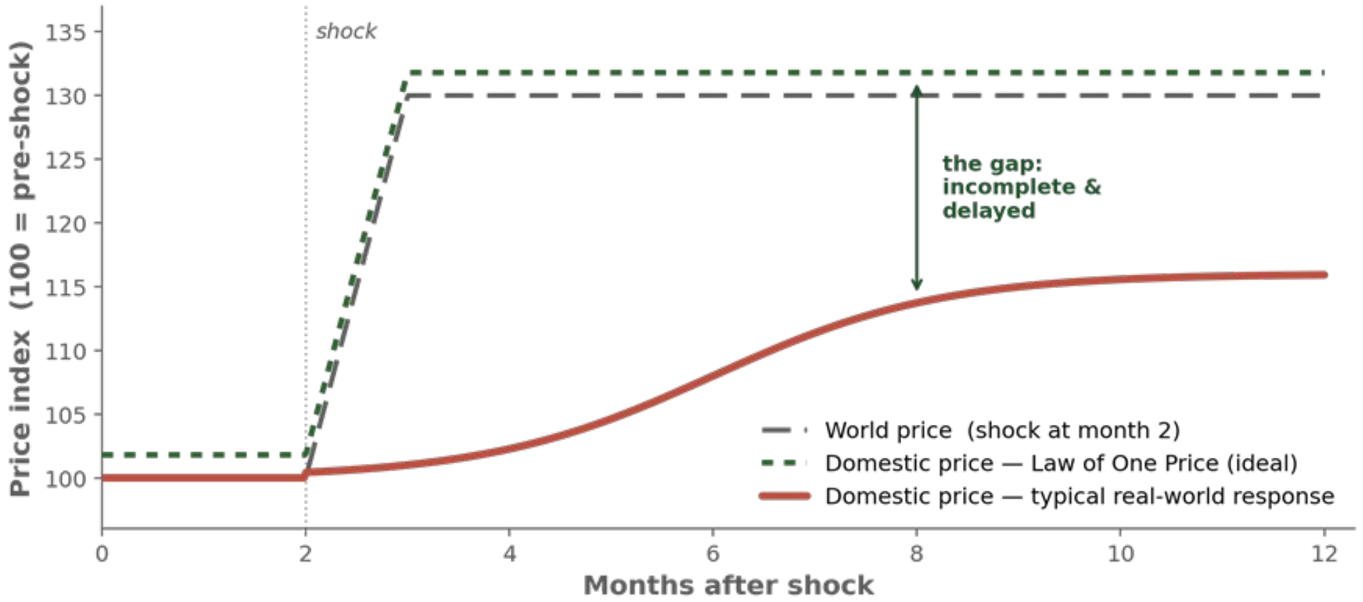
In frictionless, competitive markets, linked prices differ only by transfer costs.

LOP is the **null hypothesis** we test against.

$$P_t^D = E_t P_t^W (1 + \tau) + c$$

E = exchange rate · τ = tariff · c = transport & trade costs

Everything between P^W and P^D is the **"wedge".**



Source: Rapsomanikis et al. (2006)

Assumptions:

- **Completeness**
- **Instantaneousness**
- **Symmetry**

Sources: Rapsomanikis et al. (2006); Fackler & Goodwin (2001).

Three Dimensions of Transmission

Magnitude

How much?

$$\varepsilon = \frac{\% \Delta P^D}{\% \Delta P^W}$$

Transmission elasticity ε.

ε = 1 → full pass-through.

ε = 0.3 → only 30% passes.

Speed

How fast?



How quickly the domestic price closes the gap after a shock — days, months, or never.

Asymmetry

Same up & down?

$$\alpha^+ \neq \alpha^-$$

Do price rises and falls transmit equally? Often NOT.

Sources: Rapsomanikis et al. (2006); von Cramon-Taubadel (1998).

What Blocks or Distorts Transmission

Four forces fill the wedge between world and domestic prices:

Trade policy

→ *magnitude*

Tariffs add a wedge; export bans can even reverse transmission (India 2008).

Infrastructure & transaction costs

→ *band of no transmission*

Poor roads create a wide band where world prices move but local prices don't.

Market structure

→ *asymmetry*

A few large traders pass rises on fast, falls on slowly → asymmetry.

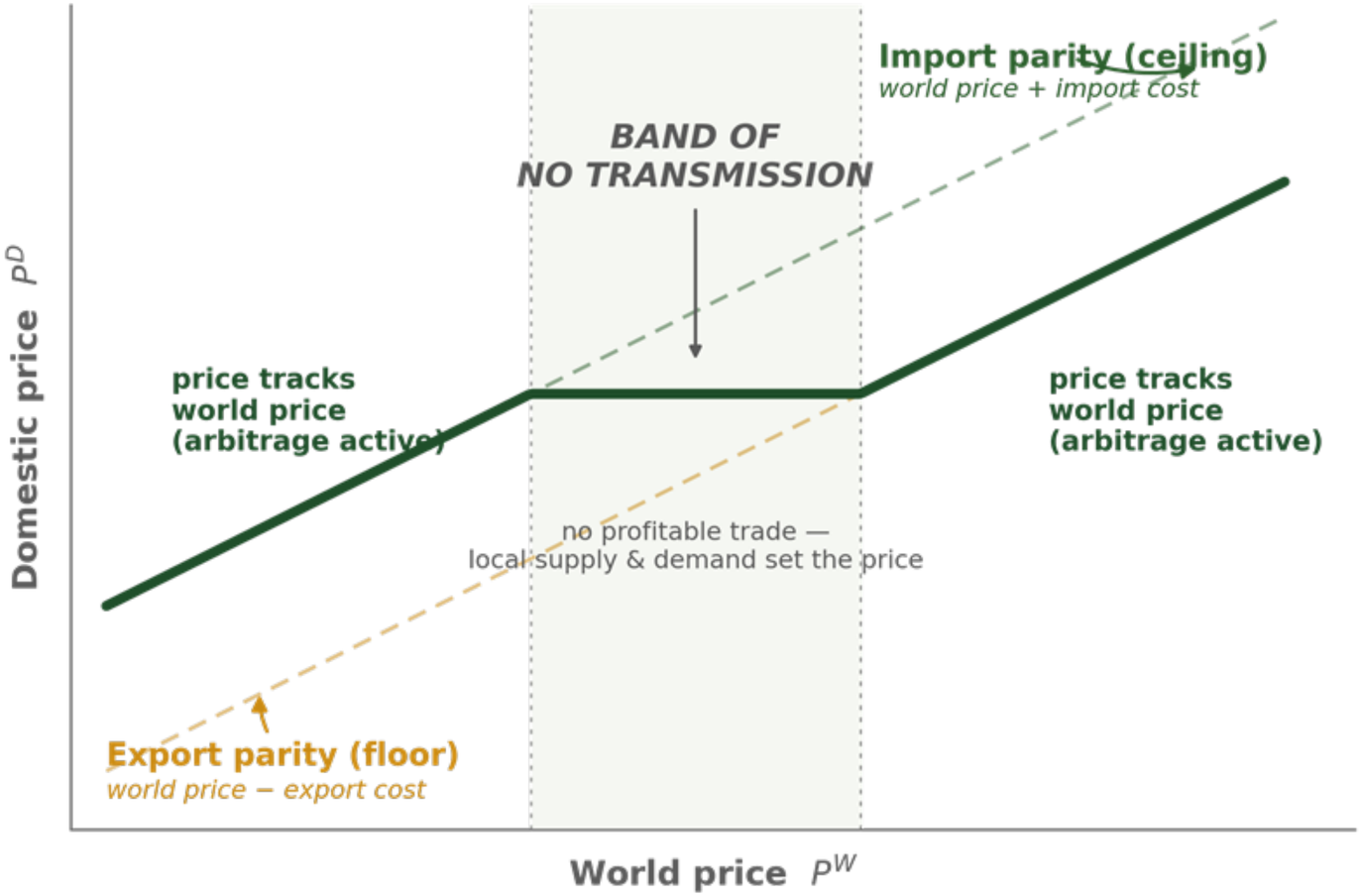
Government intervention

→ *speed*

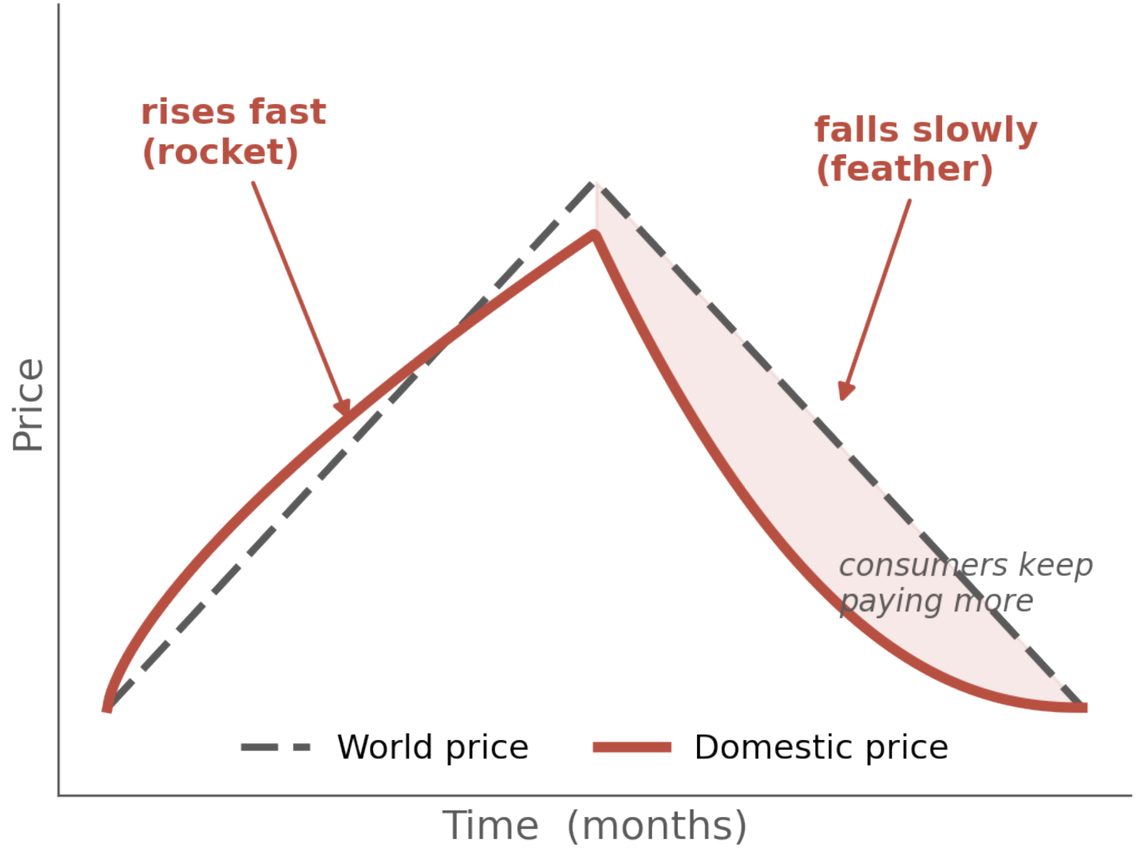
Price controls, buffer stocks, state trading deliberately decouple prices.

Sources: Minot (2011); von Cramon-Taubadel (1998); Kalkuhl (2016).

Two Key Effects, Visualised



Market power → asymmetry



Sources: Minot (2011); von Cramon-Taubadel (1998); Peltzman (2000); Abdulai (2002).

Trade Share & Country Typology

A country's trade exposure and policy stance largely determine how much transmits:

Category	Examples (developing world)	Expected transmission
Free traders	*Brazil, South Africa*	High—weak insulation
Exporting stabilisers	*China, India, Vietnam*	Low—strong policy buffers
Consistent importers	*Bangladesh, Egypt, Kenya, Senegal*	High—few stabilisation tools
Isolated economies	*Ethiopia, Malawi, Nigeria, Zambia*	Low—poorly integrated

Sources: Jensen (2013) in Pinstrup-Andersen (ed.); Minot (2011); Kalkuhl (2016).

Measuring Transmission — The Econometric Toolbox

Different questions → different methods:

Aspect of transmission	Method	
Long-run completeness (β)	Cointegration + VECM	Is there a stable world–domestic link, and how full is pass-through?
Speed of adjustment (α)	(Vector) Error-Correction Model	How fast does the gap close after a shock?
Asymmetry (rockets & feathers)	Threshold ECM (TAR/MTAR), NARDL	Do rises and falls transmit differently? (α^+ vs α^-)
Dynamics & many countries	(Panel) VAR + Impulse Responses	Trace a shock month-by-month across countries
Volatility / risk spillover	MGARCH-BEKK	Does world price uncertainty spill into local risk?

Sources: Engle & Granger (1987); Rapsomanikis et al. (2006); Enders & Granger (1998); Emediegwu & Rogna (2024).

Long-Run Link & Speed — Cointegration and VECM

1. Cointegration (long-run β)

$$P_t^D = \mu + \beta P_t^W + e_t$$

Do the two prices share a stable long-run path?

The slope β is the long-run pass-through elasticity ($\beta = 1 \rightarrow$ full transmission).

2. VECM (speed α)

$$\Delta P_t^D = \alpha (P_{t-1}^D - \beta P_{t-1}^W) + \sum_i \gamma_i \Delta P_{t-i}^D + \sum_j \delta_j \Delta P_{t-j}^W + \iota$$

α = speed of correction back to equilibrium.

≈ 0 → insulated; near 1 → fast, open market.

Speed Example:

α = -0.27

Meaning: If the world price shifts by 1 unit, the domestic price closes 27% of the remaining gap every month — so it takes about 2.2 months to close half the gap, and roughly a year to close almost all of it.

Sources: Engle & Granger (1987); Johansen (1988); Pesaran, Shin & Smith (2001).

Asymmetry, Dynamics & Volatility

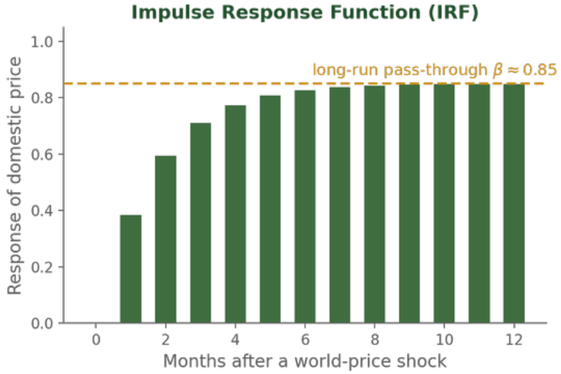
4. Asymmetry — Threshold ECM (TAR/MTAR) & NARDL

$$\Delta P_t^D = \alpha^+ ECT_{t-1}^+ + \alpha^- ECT_{t-1}^- + \ldots + u_t$$

Splits deviations into **positive** and **negative** parts to test H□: α□ = α□. Rejection = **"rockets & feathers"** (Enders & Granger 1998).

5. Dynamics — (Panel) VAR + IRF

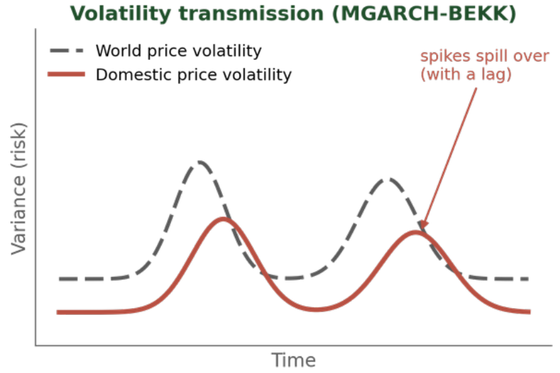
$$\mathbf{X}_t = \mathbf{c} + \sum_{i=1}^{p} \mathbf{A}_i \mathbf{X}_{t-i} + \mathbf{u}_t$$



Emediegwu & Rogna (2024) use Panel VAR across 23 countries.

6. Volatility — MGARCH-BEKK

$$\mathbf{H}_t = \mathbf{C}'\mathbf{C} + \mathbf{A}'\varepsilon_{t-1}\varepsilon_{t-1}'\mathbf{A} + \mathbf{B}'\mathbf{H}_{t-1}\mathbf{B}$$



Models risk spillover — even if price levels don't transmit, instability can.

Sources: Enders & Granger (1998); Shin et al. (2014); Emediegwu & Rogna (2024); Engle & Kroner (1995).

Illustrative Estimates (from the literature)

Country / region	Commodity	β (long-run)	α (speed)	Asymmetry?
Sub-Saharan Africa	Maize	0.31	slow	Yes (rises faster)
Bangladesh	Rice	0.72	moderate	No
Ghana	Rice (imported)	0.89	fast	Yes (rises faster)
Peru	Wheat	0.55	moderate	Mild
Ethiopia	Teff (domestic)	0.18	very slow	No (insulated)

Most current evidence — Emediegwu & Rogna (2024), Panel VAR, 23 countries:

Strong and almost immediate transmission for **maize, rice and wheat**

No significant transmission for **sorghum** — a non-traded local crop, so world prices are simply irrelevant to it.

Indicative values from Rapsomanikis et al. (2006); Minot (2011); Kalkuhl (2016). Illustrative only.

CGE Models — Capturing Economy-Wide Effects

> **β and α** measure one price link. **CGE models** instead trace a world-price shock through the whole economy.

What CGE adds

- **Distributional effects**
  producers vs consumers vs sectors
- **Macro feedback**
  GDP, exchange rate, fiscal balance
- **Economy-wide welfare**
  by household type & income group
- **Policy counterfactuals**
  tariff, export ban, cash-transfer scenarios

What drives pass-through in a CGE

Siddig & Grethe (2014):

1. **Trade share** (import / export)
2. **Sector size** in the economy
3. **Armington / CET elasticities**
   substitution: imports vs domestic goods
4. **Factor mobility** across sectors
5. **Exchange-rate regime**

Sources: Siddig & Grethe (2014), Economic Modelling 32, 212; Armington (1969).

How does domestic supply react to a world-price change?

Farmers respond to the price they *expect*—and they only ever see the **transmitted** portion of the world price.

$$\tilde{\beta} = \lambda \times \beta_S$$

λ = transmission elasticity β_s = "genuine" supply response to local price
Observed response = how much of the world signal arrives × how strongly farmers react to it.

How supply actually adjusts:

Acreage

Planted area adjusts to expected prices — but only next planting cycle.

Yield

Within-season input choices (fertiliser, labour) respond to price.

Both happen with a lag.

Sources: Nerlove (1958); Haile, Kalkuhl & von Braun (2016, in Kalkuhl et al. eds.).

Household Income and Welfare Effects

The first-order effect depends on a household's **net trade position** (Deaton 1989).

$$\frac{\Delta W_h}{Y_h} \approx \frac{\Delta Y_h^{real}}{Y_h} \approx (\phi_h^S - \phi_h^C) \times \%\Delta P$$

φ_s = income share from selling the good
φc = budget share spent buying it
$\%\Delta P$ = the transmitted domestic price change

Effect of a price RISE on real income, by household type:



Two channels of income loss for the poor:
higher food bill (large budget share 40–75%) +
forced cuts to health, schooling, assets.

Short run vs long run:
short-run poverty rises (net buyers dominate);
long-run wages & output adjust, easing the loss.

Sources: Deaton (1989); Ivanic & Martin (2008, 2014); Kalkuhl et al. (2016, eds.).

Policy Implications

High transmission

✓ Farmers get accurate price signals
→ better investment & supply response

✗ Consumers absorb the full global shock
→ food-security risk for the poor

Low transmission (insulation)

✓ Shields consumers in the short run
→ political stability during spikes

✗ Distorts incentives & efficiency losses
→ implicit tax on farmers

2007–08 lesson

When many countries insulated at once (export bans, stockpiling), each protected its own consumers but collectively pushed world prices even higher—a coordination failure that deepened the crisis.

Targeted cash transfers

preferred — protects the poor, keeps the market signal

Infrastructure investment

best long-run fix — permanently lowers the wedge

Export bans

blunt — distorts supply

Sources: Headey & Fan (2008); Martin & Anderson (2012); World Bank (2012).

Conclusions

- World price shocks pass through to developing-country food prices only partially, not fully (0.2–0.9)
- The speed of adjustment is slow and assymetric
- Both the strength and the speed of transmission depend mainly on trade exposure
- The welfare cost falls hardest on net-buyer households

Overall, world price shocks do reach developing-country markets, but only partially and with a meaningful delay.

References

- Abdulai, A. (2002). Asymmetric price transmission. Applied Economics 34(6).
- Armington, P. (1969). Demand for products by place of production. IMF Staff Papers 16(1).
- Deaton, A. (1989). Rice prices and income distribution in Thailand. Economic Journal 99.
- Emediegwu, L. & Rogna, M. (2024). Agricultural commodities' price transmission. Food Policy 126.
- Engle, R. & Granger, C. (1987). Co-integration and error correction. Econometrica 55(2).
- Fackler, P. & Goodwin, B. (2001). Spatial price analysis. Handbook of Agricultural Economics 1B.
- FAO (2011, 2014). The State of Food Insecurity in the World. Rome.
- Haile, M., Kalkuhl, M. & von Braun, J. (2016). Worldwide acreage & yield response. In Kalkuhl et al. (eds.).
- Headey, D. & Fan, S. (2008). Anatomy of a food-price crisis. Agricultural Economics 39(S1).
- Ivanic, M. & Martin, W. (2008, 2014). Food prices & poverty. Ag. Economics 39 / World Bank.
- Jensen, H. (ed., 2013). Country case studies, in Pinstrup-Andersen (ed.), Political Economy of Food Price Policy.
- Kalkuhl, M. (2016). Global commodity prices in developing countries. QJIA 55(4).
- Kalkuhl, M., von Braun, J. & Torero, M. (eds., 2016). Food Price Volatility and Its Implications. Springer.
- Martin, W. & Anderson, K. (2012). Export restrictions & price insulation. AJAE 94(2).
- Minot, N. (2011). Transmission of world food prices to Sub-Saharan Africa. IFPRI DP 01059.
- Nerlove, M. (1958). The Dynamics of Supply. Johns Hopkins University Press.
- Peltzman, S. (2000). Prices rise faster than they fall. Journal of Political Economy 108(3).
- Rapsomanikis, G., Hallam, D. & Conforti, P. (2006). Market integration. FAO Commodity Review.
- Siddig, K. & Grethe, H. (2014). Price transmission in CGE models. Economic Modelling 38: 12–22.
- von Cramon-Taubadel, S. (1998). Asymmetric price transmission. Eur. Rev. Agric. Econ. 25(1).
- World Bank (2012). Food Price Watch. Washington, DC.